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An intelligent approach for dynamic network traffic restriction using MAC address verification

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ABSTRACT

Networks are spread all throughout the world, usually found in homes, organizations, governments, etc. Due to the exponential growth in the deployment of networks, this field has become a prime target for attackers. A spoofing attack is a very common type of attack on a network. It may sound simple, but it is a complex technique involving impersonation and gaining access to restricted information. Some spoofing attacks complicate scenarios by launching other attackers using spoofed computers. Networks are usually unable to identify spoofing attacks since the hackers who attack can forge the MAC address of their device to impersonate another device on the network resulting in an advanced level of ARP spoofing thereby exposing the vulnerability of the network. This becomes really tricky when the attack is done in the address of a company's network. Several techniques exist to detect MAC spoofing such as Sequence Number Analysis, Radio Signal strength based detection, and Hop based detection. But all these techniques wrongly classify malicious user to be authenticated ones at times costing the authenticity of the valuable legitimate users. In this paper, a design of an address verification technique to detect threats by using multiplicative increase and additive decrease algorithm based on network localization with the support of Port Scanning, OS fingerprinting and Route Tracing algorithms has been discussed. This eliminates the chance of a legitimate user getting penalized during the traffic restriction resulting in a more trustworthy and a more intelligent network.

1. Introduction

Networking is the primary need for today's communication channel management and business. The growth of networks from ARPANET to today's internet is a makeable journey and a changeover from the idealization of the network to practical practices. The technical improvisation of networking infrastructure with respect to time has also lead to the misuse and wrong interpretation of the networking protocol for ill-beneficial reasons such as hacking, spoofing etc. Thus, the trivial networking infrastructure needs to be protected from malicious attackers at any given point of time. These attackers can be influencer attackers or passive attackers. The interconnected networking model provides these attackers a ground to plot and play the spoofing actions.

Despite the convenience and advantages it has, a system constituting a shared network possess not one but many malicious threats and security risks. With more number of devices connected to a system, higher is the risk of the given host system getting attacked. Yet another problem that arises with the shared network system is the intensification of traffic in a given shared network. This is why it has become highly essential to curb security threats and traffic problems in a ubiquitous network. The proposed scheme provides high constancy and compatibility because it only alters the ARP protocol. In addition, it is simple and stable, as it does not use a complex algorithm or impose extra load on the computer system.

The action of altering the MAC address is generally referred to as MAC spoofing and this can be extremely helpful for many reasons, like controlling the traffic of devices connected to a Wi-Fi hotspot. Some internet service providers bind their services to a specific MAC address; if the users change their NIC the service will not work by changing the MAC address. However, using a new interface can solve this problem. Although a large amount of research has been done in order to detect MAC address spoofing attacks in wireless networks, the existing methods including the PTD (Personal Trusted Device)-based wireless network management [1] and the wireless MAC address spoofing detection [2] are still not sufficient to discover such attacks in advance.

A progressive methodology to differentiate legitimate and spoofed user is discussed in detail here. This method ensures that all the spoofed users are identified before they start to exploit the system. Several designs of address verification technique to detect threats by implementing multiplicative increase and additive-decrease algorithm based on network localization with the support of Port Scanning,

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OS fingerprinting and Route Tracing algorithms have been discussed. MAC address validation is optimized for scaling when the router is in enhanced network services mode. Enhanced network services mode affects other features, such as multicast and firewall filters, so it is essential to take that in to consideration when deciding whether to configure enhanced mode or not.

2. Background & related works

A framework for detecting MAC and IP spoofing attacks with network characteristics that uses signal strength based detection by analysing RSSI values of packets when physical distance varies. Here Network is monitored for any changes in physical characteristics; in particular signal strength is used as a measure. The physical fingerprint of the system is also used. RSSI is a metric used to measure the received signal strength of the host, by using Euclidean distance to measure the distance of the host. It is based on the main idea that it is not that easy to mimic the physical characteristics of the network. However, physical characteristics can also be changed. The main limitation of the model is that it does not take into account if the users and the attacker are in close proximity, or if they are in exact opposite directions with the same Euclidean distance from the central host. Since it uses only signal strength as its measure to validate the physical characteristics, it fails to detect if there are no changes in signal strength; it also does not work with wired networks [2].

Detecting ARP spoofing uses the difference in captured data of MAC and IP address as a measure to decide if the ARP table is being spoofed. They inject ARP and TCP SYNC packets into the network to see if there are any malicious configured ARP Tables in the network. Though this method can be proved useful, it fails to detect if the spoof is already done and if the attacker joins the network as a user with legitimate MAC and IP, it treats the attacker as a legitimate user. This is the main drawback of the system and attackers easily manipulate this loophole to enter in to the network and pollute the ARP table [3].

Using Constant Traffic to Specific IP Destinations for Detecting Spoofed MAC Addresses in Local Area Networks addresses the issue of detecting spoofed MAC addresses in a Local Area Network, by using constant traffic to specific IP destinations as a fingerprint of the authorized station. This approach decides if an MAC is spoofed based on whether it has constant traffic to other destinations excluding what has been already recorded. This only uses Destination IP to decide if the IP is legitimate or not. But in some situations, when a user installs new software, that begins to have a different request, in which case this method registers this user as a false user, whereas the user was actually legitimate. This is the main issue against this method; it decides most users are spoofed users even though they are legitimate ones. It has a very high rate of false-positive prediction. The results are unreliable and cannot be used to accurately remove the attacker from the network [4].

Defence against Spoofed IP Traffic Using Hop-Count Filtering detects spoofing based on Hop Count. It measures the distance of the host from the central network in hop counts, based on the assumption that the attacker and user are generally not in same location. This measure fails to detect if both users and attacker are at same hop distance from the central server. And it always fails to detect one hop spoofing attacks, in which the user and the attacker are both at one hop from the central server. As one hop networks are more common in today's world, this method seems to be an age old configuration used to detect MAC spoofing attacks on the network [5,6].

Although the identity of a host can be verified through cryptographic authentication, authentication is not always possible because it requires key management and additional infrastructural overhead [7].

MAC Spoofing Detection has been considered as a main area of concern in networks as it proposes a serious threat to the network. Based on hop-count, we propose a novel filtering technique called Hop-Count Filtering to weed out spoofed IP packets at the very beginning of network processing, thus effectively protecting victim servers' resources from abuse. The rationale behind hop-count filtering (HCF) is that most randomly spoofed IP packets, when arriving at victims, do not carry hop-count values that are consistent with the IP addresses being spoofed [8].

As MAC addresses can be easily changed through device drivers, simple yet effective identity-based attacks can be implemented with off-the-shelf equipment against multiple link-layer services. IEEE 802.11 networks, for instance, have been shown to be vulnerable to a class of attacks we refer to as masquerading attacks, in which a malicious device targets a specific client by spoofing its MAC address or the address of its current access point [9].

The ability to control the functions handled by the firmware of wireless cards is being actively researched, both in an attempt to mitigate the detection techniques described in this paper and to remove the restrictions imposed by system firmware transmitting malformed management and control frames [10].

Many techniques, even detection based on physical characteristics have been tried out but it has not been possible to accurately predict if a host has been spoofed or not. Problems with this approach:

(1) Sequence Number Analysis — Can easily be changed to match the value
(2) Hop Count Filtering — Fails to work in One-Hop Networks
(3) Physical Characteristics — Pre-Dominantly use signal strength as a parameter, but if fails if both host are nearer to the host
(4) ARP Mismatch — fails to detect pre-configured attacks on the network

In [11] the author Harney et al. has patented the concept of spoofing detection in a wireless networking infrastructure. The technique is dominated on the detection of spoofing with respect to first and second time of attack under a bounded channel of wireless devices. Parallel to the approach, [12,13] a radio finger-printing technique is supported; classifying the slices of network enables the classification of the transmission with a fixed length input deep classifier. The author has also discussed the beneficiary actions and measurements in the proposed technique to avoid and detect the spoofing attacks on MAC address. The author in [14] has discussed the internal auditing survey of Address Resolution Protocol (ARP) and its dependency features towards resisting the attacks via a morphed IP or MAC address. Since the ARP is bounded to perform simpler operations, this requires an improvisation to support the activity.

Thus in relevance to networking model and framework, the author in [15] has discussed the challenges and refinement in proposing a telemedicine network under modern ICT techniques and ecosystem. The author has pointed over the networking instability to provide stable results on data transmission via intermediate channels. This is due to the weaker detection of spoofing attackers and attacking techniques. In [16] the author has extended the challenges of Internet Exchange Protocol (IXP) with respect to Border Gateway Protocol (BGP) to provide a localization support and challenging ideas towards detection and prevention of spoofing detection attacks. Thus the results are further expressed to be evaluated in near future. In [17] the Channel Fingerprint power (CFPP) technique is discussed to predict and determine the spoofing attacks on MAC address within the physical layer of operation. Making CPFF technique reliable increases the dependencies to peruse personal data in networking.

In [18] the authors have discussed a secure infrastructure to propagate the agenda of telemedicine channel under a low line networking channel. The technique aims to support the telemedicine protocol over a secure networking model with a built-in approach for malicious node/attack detection. [19] proposes the ubiquitous computing technique for detection and prediction of the MAC address based attacks. The approach is simplifying the technique by generating a monitoring unit for frequent tracking and node recording. In [20] the authors have

proposed random forest based approach to detect the MAC spoofing under received signal strength.

In [21,22], the author proposed a software based security analysis using machine learning technique. This can be done dynamically. It is also proved continues security assessment for cloud based services.

The MAC address resolution is a hot research topic with respect to the spoofing attack detection in various layers such as networking and physical. In this research survey, a key modification is proposed and designed an intelligent approach to predict and detect the spoofing attacks in MAC address.

3. Proposed work

In this paper a system is proposed, that is completely independent of physical location and characteristic of the system. We use a threshold based system, so that even if a false prediction has been made it can be corrected before classifying it as an attacker. We also use a self-learning model so that for each network it is deployed, it learns the network's characteristics on its own, enabling it to predict the spoofing faster. In [22], risk assessment frameworks were designed to identify risks based on previously obtained data and experience. The self-learning system proposed enables it to anticipate and predict risks. This is based on the host characteristics and so avoids false accuses of a host. We try to punish a host for a wrong value but we give them a chance to have it stable, by decreasing the threshold once the value gets stable. We increase the threshold exponentially for each wrong projection and decrease it linearly when it becomes stable, thus, the technique is able to detect attacker hosts much more accurately without wrongly accusing the legitimate hosts. By this progressive system we avoid detecting false positives and wrongly accusing a legitimate user. This system also prevents the attackers from exploiting the system; the more stable a network is, the faster are the chances of finding out attackers. If the network is unstable, it takes more time to detect attackers in the network giving them access to exploit and manipulate.

3.1. System architecture

The central server acts as hotspot for devices to connect to the network, it is also used for scanning the host systems and updating the local data of the network. When a new host joins the network it sends a ARP request asking who is at this IP address. Host responds with a ARP response saying its MAC address is at this IP address. When the attacker uses the same MAC address, it gives out the same IP address to the attacker thereby spoofing the network, Now the scanning starts and our algorithm views the scanned reports and specifies the user a legitimate or not based his current threshold for his host in the system (see Fig. 1).

4. Mathematical model & algorithms

4.1. Proof of concept

Consider the Network (N) with an incoming request as $req(n)$ such that, $n \rightarrow 1, 2, 3 \ldots$ for every value of (n), the request fetches the port quadrant attributes such as IP address of source (S_{IP}), Destination IP (D_{IP}), Data Frame (F_D) and Mapping Route Addresses (M_R). Thus on every request, $req(n)$, there implies a pairing as in Eq. (1).

$$P_D = \int_0^n \frac{req\,(n)}{F_D} \tag{1}$$

$$P_D = \int_0^n \sum_{n=1}^{\infty} \frac{req(n)}{F_D} \left(D_{IP}\right) \tag{2}$$

Similarly, the Eq. (2) can be represented as in Eq. (3) for IP detection in sender

$$P_S = \int_0^n \sum_{n=1}^{\infty} \frac{req(n)}{F_D} \left(S_{IP}\right) \tag{3}$$

Thus, each of the pair of attributes are stored and reported to trusted authority (TA) and thus TA can be represented as TA = $\sum P_S + P_D$, Hence on the detection of spoofing attacker or malicious node, the TA summarization value is different with respect to the threshold value. Thus algorithm, (A_1) and (A_2) represents the stage of evaluation.

Hence, on malicious node/user prediction, the TA can be represented as in Eq. (4)

$$T_A = \int_0^2 \sum_{n=1}^{\infty} \frac{d(req\,(n))}{d(t)} \left(D_{IP}\right)\left(S_{IP}\right) \tag{4}$$

Where, the function represents the MAC address validation as shown in Section 4.2.

4.2. Multiplicative increase and additive decrease

The Central host sends out messages on the network and produces the result. Now our algorithm compares these values with the previous values and decides if there is a mismatch in metrics. It increases its threshold if there is a mismatch in configuration to punish the system for inconsistent information, also if same value is returned, to reward the system the threshold is decreased. So ultimately when the threshold goes above the max threshold the system is said to have spoofed the MAC address.

Phases

1. Increase threshold by a factor of 2 if mismatch in result.
2. Decrease threshold linearly if exact match of mismatched result.

Pseudo codes:

Pseudo code 1: For Port
Begin
Input
If port is equal to current type **then**
    Print Port already exists
    End Execution
Else
    If port equals to the theMisMatchvalue of the current type **then**
        Decrease threshold linearly
        Set the current port with updated values
    Else
        Increase the threshold by multiplying it with 2
        Set the current port with the updated values

    End If
End If
End
Output: Updated port value/threshold

Pseudo code 2: ForTraceRoute
Begin
Input trace route (if exist)
Iftrace route already exist **then**
    Select the existing route
Else
    Create a new trace route
    Re route all packets to this route
End If
End
Output: A specific trace route

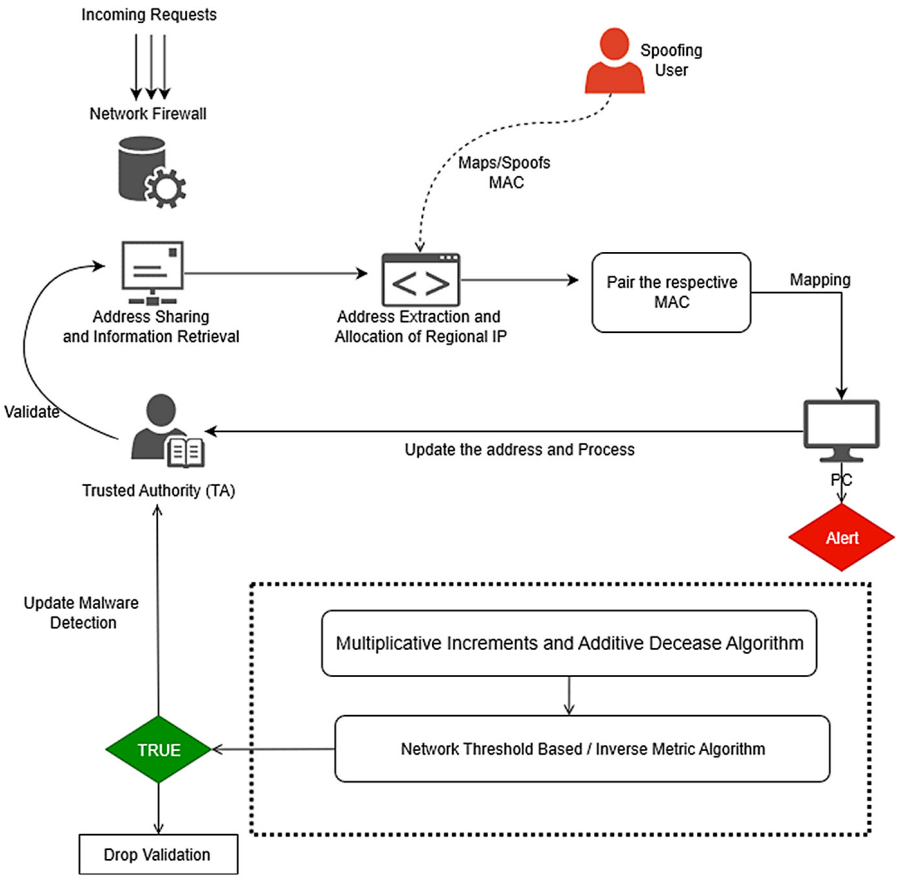


Fig. 1. System architecture.

Pseudocode 3: For BootTime
Input: bootTime(if exist)
Begin
If bootTime is null **then**
Create a new entry of bootTime
Else
Update the existing bootTime to current Time
End If
End
Output : Updated bootTime

Pseudo code 4: For OS
Input: OS entry(if exist)
Begin
If there is not entry in OS **then**
New Entry OS
Else If OS entry value equals to the existing OS value
Already existing OS
Decrement OS
Else
Set the OS value to be current OS
Increment OS value
End If
End
Output: Updated OS value

5. Network threshold based on inverse metric algorithm

1. Update global metrics on joining of new host.
2. Updating metrics

Step 1: Compute the variations
Step 2: If less variation threshold is high
Step 3: If variation is high threshold is less

This is used to localize the parameters of the local network and decide the increase factor of threshold based on the global values for the local network. If the network characteristics are more stable around the same value, the threshold for it is very high; if it is not stable and highly various for each system, it is set to a lower value. This technique is ideally used to create a dynamic threshold configuration for each local network it acts on thereby providing optimal performance on each network. Since the same threshold becomes ineffective in various networks, proposed system is designed to self-learn the threshold values based on the host characteristics in the system [23].

6. Calcuations and result

Compared to previous techniques this technique provides a progressive way to detect if an attacker has spoofed into the network. All the previous techniques had the possibility of making false predictions, thereby predicting a legitimate user as spoofed one, whereas our algorithm learns the network, decides the optimal threshold for the hyper parameters of the system and progressively finds out or eliminates the attacker without producing any false-positive predictions. Our Algorithm outperforms the entire existing algorithm in terms of accuracy of false predictions while at same time not giving up on legitimate predictions. The overall false prediction accuracy is less than

Table 5.1
Sample calculations for algorithm 4A: multiplicative increase and addictive decrease (Legitimate user).

User	Port value	Threshold for port value	bootTime	OS entry	Status
192.168.2.24	20 400	20 400	18:02:20	New	Existing
192.168.2.25	9 880	10 000	18:02:20	Increment OS	Increment
192.168.2.25	11 789	20 000	18:02:21	Decrement OS	MisMatchtype:Decrement
192.168.2.25	12 789	19 920	18:02:22	Decrement OS	Decrement
192.168.2.25	13 678	19 690	18:02:23	Decrement OS	Decrement
192.168.2.25	16 689	19 360	18:02:24	Decrement OS	Decrement
192.168.2.25	13 569	19 080	18:02:25	Decrement OS	Decrement
192.168.2.25	12 454	18 760	18:02:26	Decrement OS	Decrement
192.168.2.25	13 567	18 490	18:02:27	Decrement OS	Decrement

Table 5.2
Sample calculations for Algorithm 4A: multiplicative increase and addictive decrease (Attacker).

User(attacker)	Port value	Threshold for port value	bootTime	OS entry	Status
192.168.3.24	20 400	20 400	18:02:20	New	Existing
192.168.3.27	9 880	10 000	18:02:20	Increment OS	Increment
192.168.3.27	20 911	20 000	18:02:21	Increment OS	Increment
192.168.3.27	31 910	40 000	18:02:22	Decrement OS	MisMatchtype:Decrement
192.168.3.27	41 189	39 720	18:02:23	Increment OS	Increment
192.168.3.27	52 456	79 440	18:02:24	Decrement OS	Decrement
192.168.3.27	68 900	79 120	18:02:25	Decrement OS	Decrement
192.168.3.27	89 456	78 740	18:02:26	Increment OS	Increment
192.168.3.27	99 897	157 480	18:02:27	Attacker	Reject packets

3% calculated based on stimulation performed using various levels of networks. Table 5.1 gives an idea of the working of Algorithm 1.

Clearly the Table 5.1 shows that the user is legitimate as the threshold values adhere with the specified values, even if one of the entries is out of bounds, it is not penalized. This is the proof that the algorithm does not penalize the legitimate users even if there is a slight change in values because of the network issues. Sometimes network congestion leads to loss of packets and the actual port value may be very less than the specified threshold. In such extraordinary cases, the user is considered to be worthy of another few chances before his packets are dropped completely. Now let us consider a table with values from an attacker trying to break threshold (see Table 5.2).

Result graph

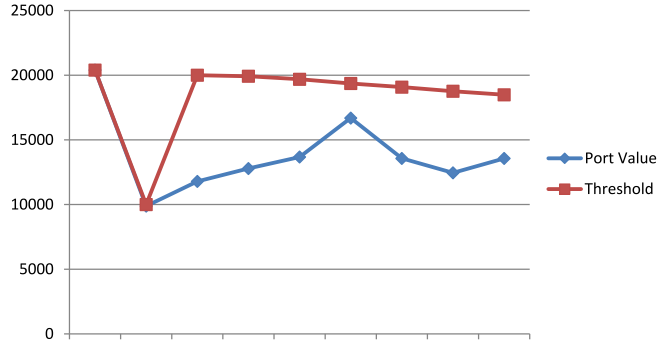
In this session, the proposed approach is evaluated with respect to the users and attack nature, i.e. the legitimate users and attackers; the proposed system proves the dependencies of MAC address using ARP in the initial stage with the maximum threshold of resolving or resisting the external mitigation attacks. Thus, the overall system is aided with an improvised approach as discussed below.

The following graph shows the comparison between two users:

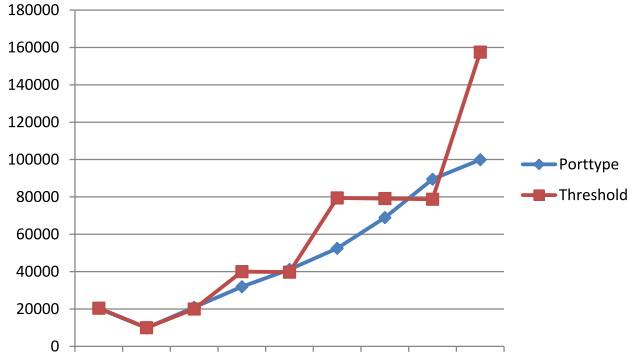
(a) Legitimate user
(b) Attacker

Graph 5a clearly demonstrates the port values and the threshold numbers that decrements slowly and gives chance for the legitimate user rather than penalizing it. This is the working proof of the algorithm and the most important aspect of the work.

Graph 5b clearly demonstrates the port-Type values and the threshold values for the attacker. There is a clear evidence of punishing the attacker and multiplicative Increment makes sure that the attacker's plan of compromising the system or its resources is kept at bay.



Graph 5a. Legitimate user.



Graph 5b. Attacker.

7. Conclusion and future work

New threats to data security in a network keep emerging almost every day. Even though these threats cannot be premeditated, they can be seamlessly eliminated by using AI-enabled machine learning that looks out for more hyper parameters in the security-check process. By doing so, we not only protect the data from the threat but also make the system aware of any such attacks in the future. This proves to be extremely useful in an Industrial scenario, where prior elimination of a threat in a software-defined networking system ensures even more optimized speed for unstable networks where the threshold value computed is lesser than normal.

This benefits both the client and the customer trying to get input and output values of data being entered and stored in the system. By leveraging the AI with information about the MAC address of the system in use and the attacking system, the need of human intervention is completely eliminated and the network becomes even more secure

against proliferating attacks by highly-skilled hackers. When the traffic in the network sees a rapid rise, even if it does not create any new threats, it leads to an overall increase in the network bandwidth of the entire system which will now be equal to the sum of bandwidth of the original network and the various traffic networks present. By using AI to dynamically control the traffics in a given network even these problems of increased bandwidth can be entirely eradicated or optimized.

Whenever an AI is used for scanning a network for possible threats, the defender as well as the attacker become aware of the various scans involved. The reason why the attacker is also aware is because it is easy for him to notice changes in the host as he continuously supplies traffic to his host. In our scenario the AI that is used for identifying the MAC address of scenario can be disguised as just another extended feature of the network and by doing this it does not raise the suspicion of the attacker, who without any doubt will now fall for the bait. Proposed strategy of using MAC address verification along with a fully self-empowered AI can do revolutions in networking that we could have never possibly imagined, but the greatest revolution out of it would be the tremendous conservation of time at both the ends of the network that the removal of traffic would bring.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

M. Anathi: Writing - original draft, Visualization, Investigation, Validation. **K. Vijayakumar:** Conceptualization, Methodology, Software, Writing - review & editing.

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